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Exp - 9

Dry Run :

- initialize array ⇒ arr = [1, 6, 11, 5]

Step 1 : Calculate Total Sum :

totSum = 1+6+11+5 = 23

Step 2 : Initialize DP Array :

dp Size = 24 (0 to 23)

Initially :

dp[0] = true

rest all = false

i.e, dp = [T, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F]

Step 3 : Process Each Element :

- Take num = 1

Update from j = 23 → 1

Only change :

dp[1] = dp[1] OR dp[0] = true

i.e, dp = [T, T, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F]

- Take num = 6

Update from j = 23 → 6

Changes : dp[6] = dp[6] OR dp[0] = True

→ dp[7] = dp[7] OR dp[1] = T

dp = [T, T, F, F, F, F, T, T, F, F, F, F, F, F, F, F, F, F, F, F, F, F, F]

- Take num = 11

  Update from j = 23 → 11

  Changes: dp[11] = dp[11] OR dp[0] = T

  dp[12] = dp[12] OR dp[1] = T

  dp[17] = dp[17] OR dp[6] = T

  dp[18] = dp[18] OR dp[7] = T

∴ dp = [T, T, F, F, F, F, T, T, F, F, F, T, T, F, F, F, F, T, T, F, F, F, F, F]

- Take num = 5

  Update from j = 23 → 5

  Changes: dp[5] = dp[5] OR dp[0] = T

  dp[6] = already T

  dp[7] = already T

  dp[10] = dp[10] OR dp[5] = T

  dp[11] = already T

  dp[12] = already T

  dp[16] = dp[16] OR dp[11] = T

  dp[17] = already T

  dp[18] = already T

  dp[22] = dp[22] OR dp[17] = T

  dp[23] = dp[23] OR dp[18] = T

<u>Final dp</u>: dp = [T, T, F, F, F, T, T, T, F, F, T, T, T, F, F, F, T, T, T, F, F, F, T, T]

<u>Step 4</u>: <u>Find Minimum Difference</u>:

We check from j = 0 → 11 (since totSum/2 = 11)

Valid sum (where dp[j] = true):

j = 0, 1, 5, 6, 7, 10, 11

Now Computing;

diff = totSum - 2* j

j	diff = 23-2* j
0	23
1	21
5	13
6	11
7	9
10	3
11	1 ☑ (minimum)

<u>Final</u> <u>Answer</u>:

Minimum Difference = 1